



UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Master's Degree in Data Science and Business
Informatics

Business Process Modelling
Project Report
[P54: Scuola di recitazione]

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Capitolo 1

Introduction

1.1 Introduction

This report describes the modeling and analysis of a business process for managing student requests in an acting school. The main goal is to design a process that shows how the student, the teacher, and the school office interact from the first contact to the end of the course. The process covers the full student experience: the student contacts the school for information about available courses, selects a course, and is assigned a teacher. It also includes scheduling lessons, running the training sessions, and handling administrative tasks such as electronic payments. The model considers situations where exercises may need to be repeated, additional lessons are scheduled, or the student decides to end the course. The modeling is done in several steps. First, the workflows are drawn using Business Process Model and Notation (BPMN), giving a clear visual view of the process. The system is shown as a BPMN collaboration with two pools: the Student and the School. The School pool is divided into lanes representing the teacher and the school office. BPMN diagrams were created with Camunda Modeler. Next, each BPMN model is converted into a formal Workflow Net (WfNet) based on Petri net theory. This allows checking the structure and behavior of the processes. WoPeD is used to verify important properties like boundedness and liveness. The report ends by discussing the soundness results for both the individual participant processes and the overall collaboration model.

Capitolo 2

BPMN diagram

2.1 BPMN diagram

For the initial stage of process design, Business Process Model and Notation (BPMN) was selected as the modeling standard. BPMN provides a graphical representation of business workflows, making them accessible to both technical and non-technical stakeholders. The clarity of the notation and the user-friendly interface of the Camunda Modeler are particularly well suited for capturing the complexity of the considered scenario, including the interactions between the student and the school. Figure 2.1 presents the BPMN collaboration diagram representing the overall interaction between the two main participants: the Student and the Acting School. Each participant is modeled as a separate pool, and the interaction between them is represented through message flows. The following sections describe in detail the internal workflows of each participant through their respective process diagrams.

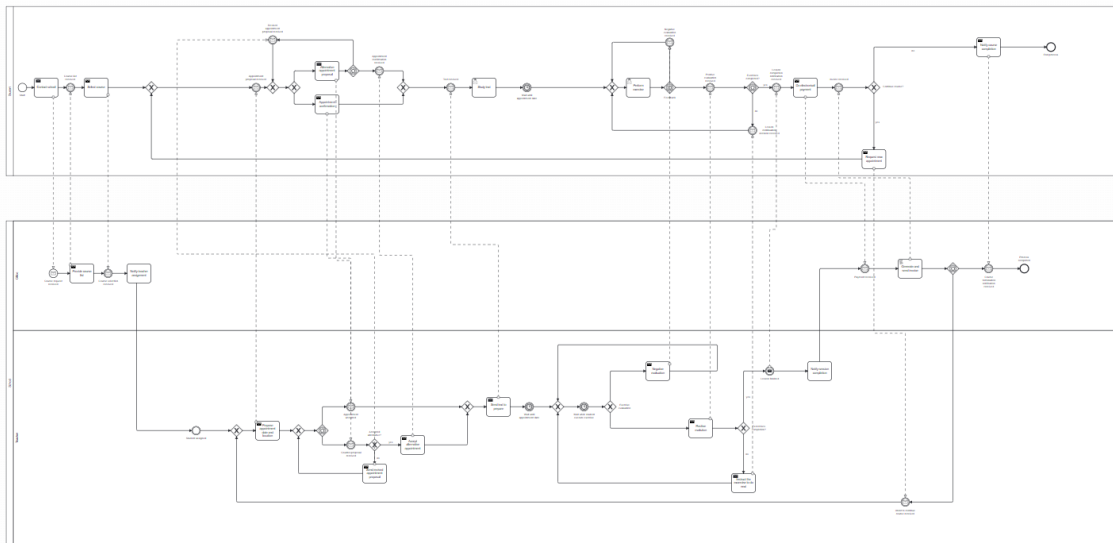


Figure 2.1: Full Acting School Collaboration BPMN

2.1.1 Student

The Student pool (Figure 2.2) represents the workflow from the student’s perspective within the acting school scenario. The process starts with a Start Event, which models the student’s decision to contact the acting school in order to begin a training program. The first activity is the Send Task “Contact School”, through which the student sends a request to the school. This is followed by a Message Catch Event, representing the reception of the list of available courses. The student then performs the Send Task “Select Course”, choosing the desired course. The process then enters the appointment negotiation phase. The student receives a proposal for date and location from the teacher via the Message Catch Event “Appointment Proposal Received”. The evaluation of this proposal is modeled through an Exclusive Gateway (XOR). If the proposal is accepted, the student performs the Send Task “Confirm Appointment”. Otherwise, the student performs the Send Task “Send Alternative Appointment Proposal”, suggesting a new date and location. This interaction forms a loop that continues until an agreement is reached. Once the appointment is confirmed, the student receives the text to prepare through a Message Catch Event and performs the User Task “Study Text”. Then, a Timer Intermediate Event (“Wait Until Appointment Date”) models the waiting time before the lesson. During the lesson, the student executes the User Task “Perform Exercise”. After completing the exercise, the process waits for feedback from the teacher through an Event-Based Gateway, which leads to two possible Message Catch Events: “Negative Evaluation Received” or “Positive Evaluation Received”. In case of negative evaluation, the flow loops back to the exercise execution. In case of positive evaluation, the process proceeds to verify whether additional exercises are required. An Event-Based Gateway (“Exercise Completed?”) manages the continuation of the exercises. If the lesson is not finished, the flow returns to another exercise; otherwise, the student receives, through a Message Catch Event, the notification from the teacher that the lesson has finished, and then the process proceeds to the payment phase. At the end of the lesson, the student performs the Send Task “Do Electronic Payment”, followed by a Message Catch Event representing the payment confirmation from the school. Finally, an Exclusive Gateway (“Continue Course?”) determines whether the student continues the course or terminates it. If the student decides to continue, the process returns to the appointment scheduling phase. Otherwise, the student performs the Send Task “Notify Course Completion”, after which the process ends.

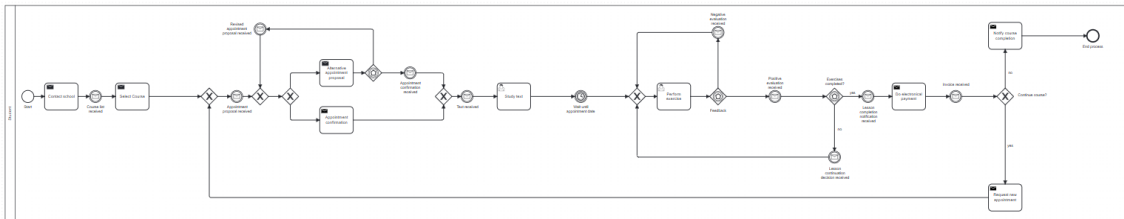


Figura 2.2: Student pool BPMN

2.1.2 School

The School pool (Figure 2.3) represents a process modeled within a single pool, divided into multiple lanes corresponding to the roles involved in the workflow, such as the administrative office and the teacher. The process begins with a Message Start Event triggered when a course request is received from the student. In the office lane, the initial activities are performed: the request is read, and the office sends the course list to the student. Upon receiving the selected course through a Message Catch Event, the office notifies the teacher responsible for that course. Next, the process moves to the teacher lane, where the teacher manages the appointment scheduling phase. The teacher proposes a date and location for the lesson and waits for the student's response through an Event-Based Gateway. This interaction, modeled as message exchanges, may repeat until an agreement is reached. Once the appointment is confirmed, the teacher sends a text message to the student with preparation instructions. After an intermediate timer event, the lesson phase begins. During the lesson, the teacher oversees the execution of exercises and evaluates the student's performance. Based on the evaluation, the process may require the student to repeat the exercise or allow the workflow to proceed. These alternative paths are controlled through exclusive gateways, which determine the appropriate continuation of the process. After completing the lesson activities, the teacher informs the student of the lesson's end and notifies the office of the session's completion. The process then moves to the payment stage, where the school receives the electronic payment from the student and issues the corresponding invoice. At the end of each lesson, after the electronic payment, the student and the teacher may agree on a new appointment following the procedure described above, or the student may decide to terminate the course. If the course continues, the Event-Based Gateway in the office lane directs the flow to the teacher lane via a Message Catch Event labeled 'Intent to Continue the Course.' If the course terminates, the gateway remains in the office lane, where the office receives the student's termination notice through a Message Catch Event, after which the process ends.

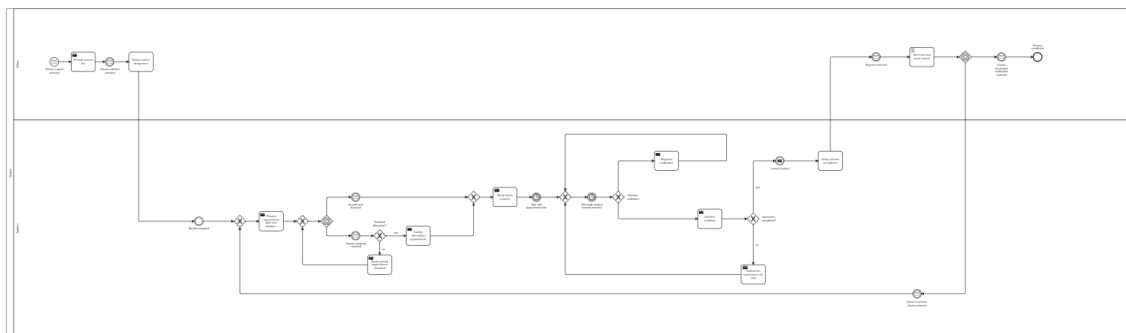


Figura 2.3: School pool BPMN

Capitolo 3

Petri Net

3.1 Petri Net

The BPMN processes were then transformed into Workflow Nets, a specific type of Petri net. Using Petri nets provides a more formal and mathematical representation of the process, allowing the verification of properties such as soundness, liveness, and boundedness.

The conversion followed these rules:

- Each task and event was replaced by a transition.
- Each sequence flow was replaced by a place.
- XOR splits and XOR joins were modeled using places and transitions to ensure mutual exclusivity.
- Event-based gateways were translated into a place followed by multiple transitions representing the possible alternatives.
- Message flows were modeled by connecting sending and receiving activities through dedicated places.
- A single initial place and a single final place were added.

The Workflow Nets for the individual processes (Student and School pools) were generated using the BPMN-to-Petri Net tool. The Petri net representing the full collaboration process was instead created manually following the same rules. The resulting nets were analyzed using the WoPeD tool to verify their structural and behavioral properties. Coverability graphs were also generated and are reported in the Appendix.

3.1.1 Student

The workflow net for the student process is shown in Figure 3.1 and consists of 32 places, 36 transitions, and 72 arcs. From the structural analysis, it emerges that

the net satisfies the workflow net property, with a single input place and a single output place, and every place and transition lies on at least one path from the input to the output. Examining the pre-set and post-set of the transitions, it is observed that each transition has exactly one incoming and one outgoing arc. This structural property allows the net to be classified as an S-net. The network also satisfies the free-choice property, since for each pair of transitions the corresponding pre-sets are either disjoint or identical. This ensures that any conflicts are local and well-structured, preventing structural deadlocks and enabling tokens to progress along valid paths. The S-component analysis identifies a single component covering all places and transitions of the net, indicating that the extended network forms a strongly connected structure. Together with the absence of free-choice violations, these structural properties contribute to the correct behavioral characteristics of the model. The analysis performed with WoPeD confirms that the net is bounded, live, safe, and sound. The coverability graph was generated using WoPeD to further validate these properties. Due to the boundedness of the net, the coverability graph coincides with the reachability graph, providing a finite representation of all reachable markings. Overall, the student workflow net is bounded, live, safe, sound, and well-structured, demonstrating that the modeled process behaves correctly and reliably.

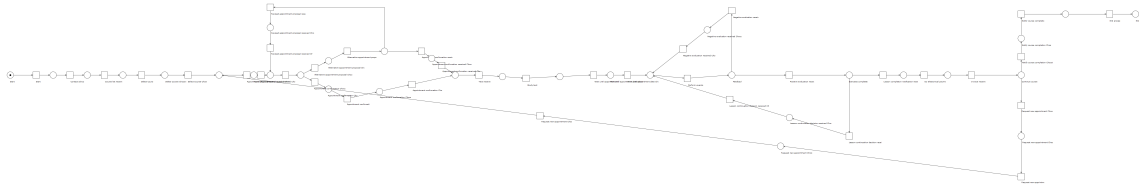


Figura 3.1: Student WF net

3.1.2 School

The workflow net for the school process (office–teacher) is shown in Figure 3.2 and consists of 35 places, 39 transitions, and 78 arcs. The structural analysis confirms that the model satisfies the workflow net property, with a unique input and output place and with every node belonging to a path from the input to the output. Similarly to the student net, the examination of the pre-set and post-set of each transition shows that every transition has exactly one input and one output place, allowing the network to be classified as an S-net. The model also satisfies the free-choice property and the S-component analysis identifies a single component covering the entire net, indicating a strongly connected structure. The behavioral analysis performed with WoPeD confirms that the network is bounded, live, safe, and sound. As in the previous case, the coverability graph coincides with the reachability graph due to the bounded nature of the net, providing a finite representation of all reachable markings. Overall, the School workflow net exhibits the same structural

and behavioral properties as the student net and therefore represents a correct and reliable process model.

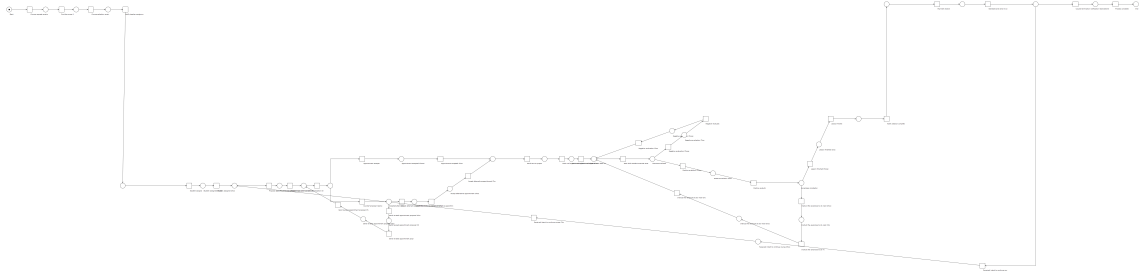


Figura 3.2: School WF net

3.1.3 Full process

The complete workflow net representing the interaction between the Student and the School processes consists of 51 places, 43 transitions, and 120 arcs, as shown in Figure 3.3. This network represents the full collaboration model obtained by integrating the individual participant processes. Structural analysis highlights some differences compared to the isolated workflow nets. Five free-choice violations are present, reflecting the interactions and dependencies between the two participants. The S-component analysis identifies 210 S-components; however, all places remain covered by at least one component, ensuring that tokens circulate within connected sub-networks. The well-structuredness check indicates the presence of 150 PT-handles and 136 TP-handles, confirming that the network is not fully well-structured. Such structural complexity is typical when combining multiple participant processes into a single collaboration net. Despite these structural characteristics, behavioral analysis confirms that the workflow net is sound. The workflow net property and initial marking are satisfied. Boundedness analysis shows that no place is unbounded, and liveness analysis indicates that all transitions are live with no dead transitions. Therefore, even though the collaboration network is not well-structured and includes multiple free-choice violations, it remains bounded, live, and sound, demonstrating that the overall interaction between the student and the school processes is correctly modeled.

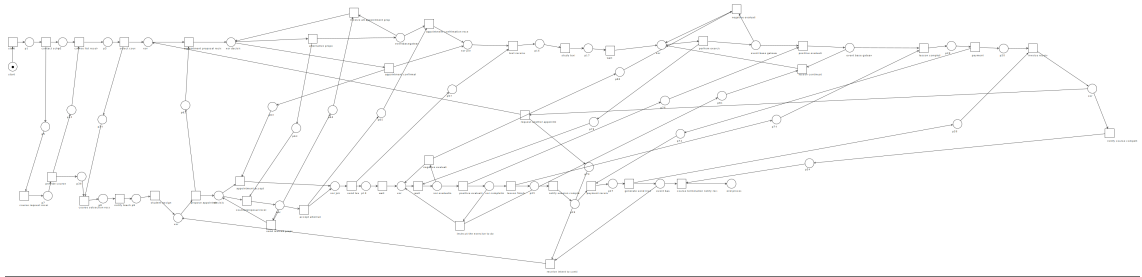


Figura 3.3: Collaboration WF net

Appendice A

Appendix

A.1 Some other Workflow Net image

A.1.1 Workflow nets statistics

Net statistics

Places:	32
Transitions:	36
- Operators:	0
- Subprocesses:	0
Arcs:	72

Figura A.1: Student statistics

Net statistics

Places:	35
Transitions:	39
- Operators:	0
- Subprocesses:	0
Arcs:	78

Figura A.2: School statistics

Net statistics

Places:	51
Transitions:	43
- Operators:	0
- Subprocesses:	0
Arcs:	120

Figura A.3: Full process statistics

A.1.2 Workflow nets analysis

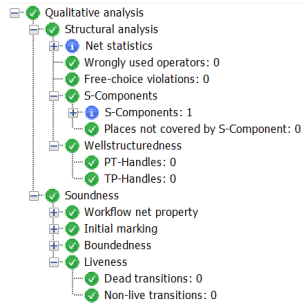


Figure A.4: Student analysis

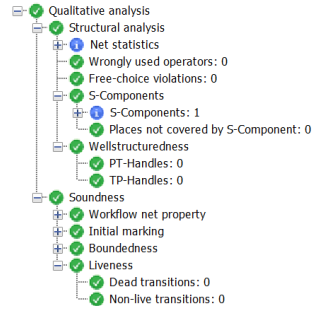


Figure A.5: School analysis

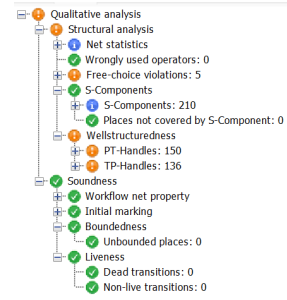


Figure A.6: Full process analysis

A.1.3 Workflow nets coverability graphs

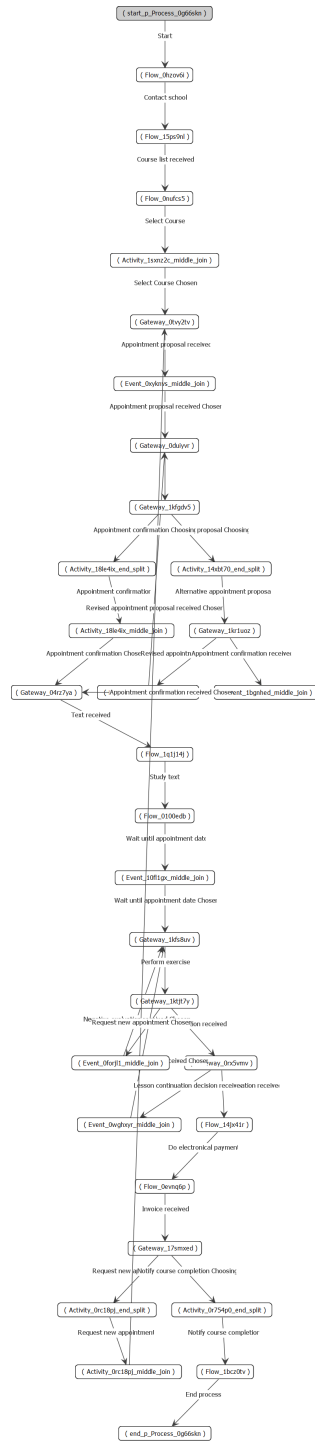


Figura A.7: Student coverability



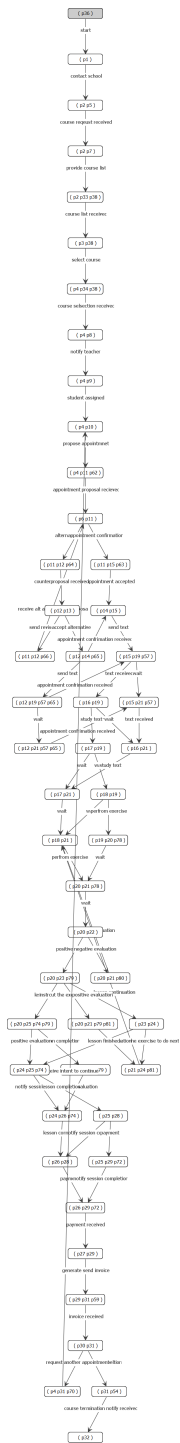


Figura A.9: Full process coverability